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embeddings → representation of data

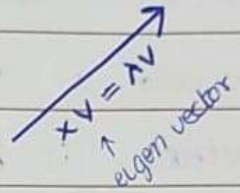
$\lambda \to$ square matrix

$\sigma \to$ any matrix

# AUTOENCODERS ⟷ Representation Learning

intro to

**Recap:** EVD




X → Eigen Value Decomposition

square

Singular value decomposition

$$X = U\Sigma V^T \to \begin{bmatrix} \sigma_1 & \cdots & 0 \\ 0 & & \sigma_n \end{bmatrix}$$

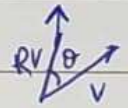
• $X = U \lambda U^T$

$$\downarrow$$

$$\begin{bmatrix} \lambda_1 & & & & 0 \\ & \lambda_2 & & & \\ & & \lambda_3 & & \\ & & & \ddots & \\ 0 & & & & \lambda_n \end{bmatrix}$$

• Rotation Matrix

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$




Deep Learning & EVD → Reasoning

| X | → | X |
|---|---|---|
| 10,000 × 100 | | 10k×50 |

- 10,000 ↓ no. of samples
- 100 ↓ # of features
- 10k×50 ↓ imp. features (as of spread of data)

compression of data

data in form of eigen values

$$\begin{bmatrix} 20 & & & & & & \\ & 10 & & & & & \\ & & 2 & & & & \\ & & & 1 & & & \\ & & & & 0.1 & & \\ & & & & & 0.01 & \\ & & & & & & 00 \end{bmatrix}$$

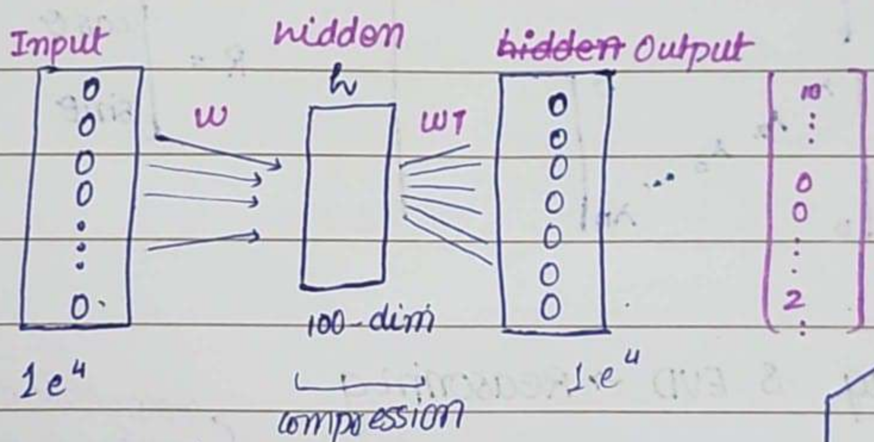
- Autoencoders

unsupervised neural networks in deep learning that learn to compress data into a lower-dimensional representation (encoding) and the reconstruct it back into orignal form (decoding)

Representation:

$$x \longrightarrow \boxed{\text{Encoder}} \longrightarrow \boxed{h} \longrightarrow \boxed{\text{Decoder}} \longrightarrow \hat{x}$$

$$L = \|x - \hat{x}\|_2^2 \longrightarrow \text{Reconstruction loss}$$




need:
lots of zeros
→ Sparsity

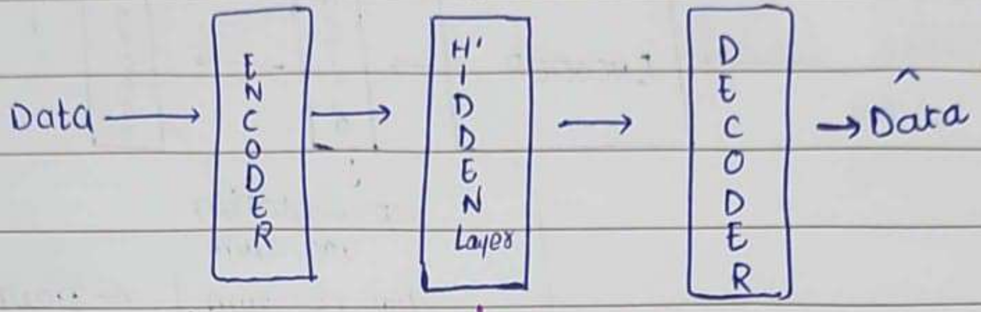
for this
L1-Norm will be introduced

$$L = \|X - \hat{X}\|_2^2 + \alpha\, L\text{-}1\text{Norm}$$

SPARSE AUTOENCODERS

## AUTOENCODERS:

Bottle Neck




Latent Layer

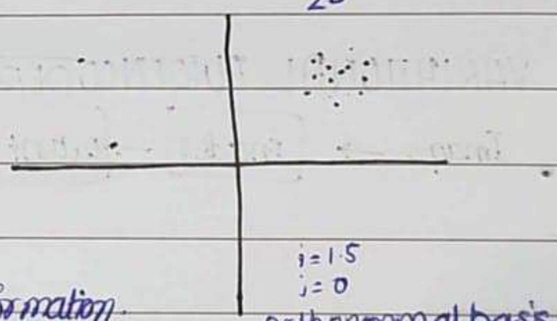
$X = Q \lambda Q^T \longrightarrow$ EVD

$X = U \Sigma V^T \longrightarrow$ SVD

$n \times p$ — n: no. of samples, p: dimension

orthonormal vectors

transformation




i = 1.5
j = 0

orthonormal basis

$\begin{bmatrix} \cancel{0}1 & 0 \\ 0 & 1 \end{bmatrix}$

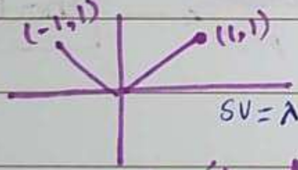
$\begin{bmatrix} 0 & 1.5 \\ ij & = 0 \end{bmatrix}$

**Example:**

$A = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$




$SV = \lambda V$

scaler $S = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$ doesn't change ~~brings change~~ in direction

Linear Transformation

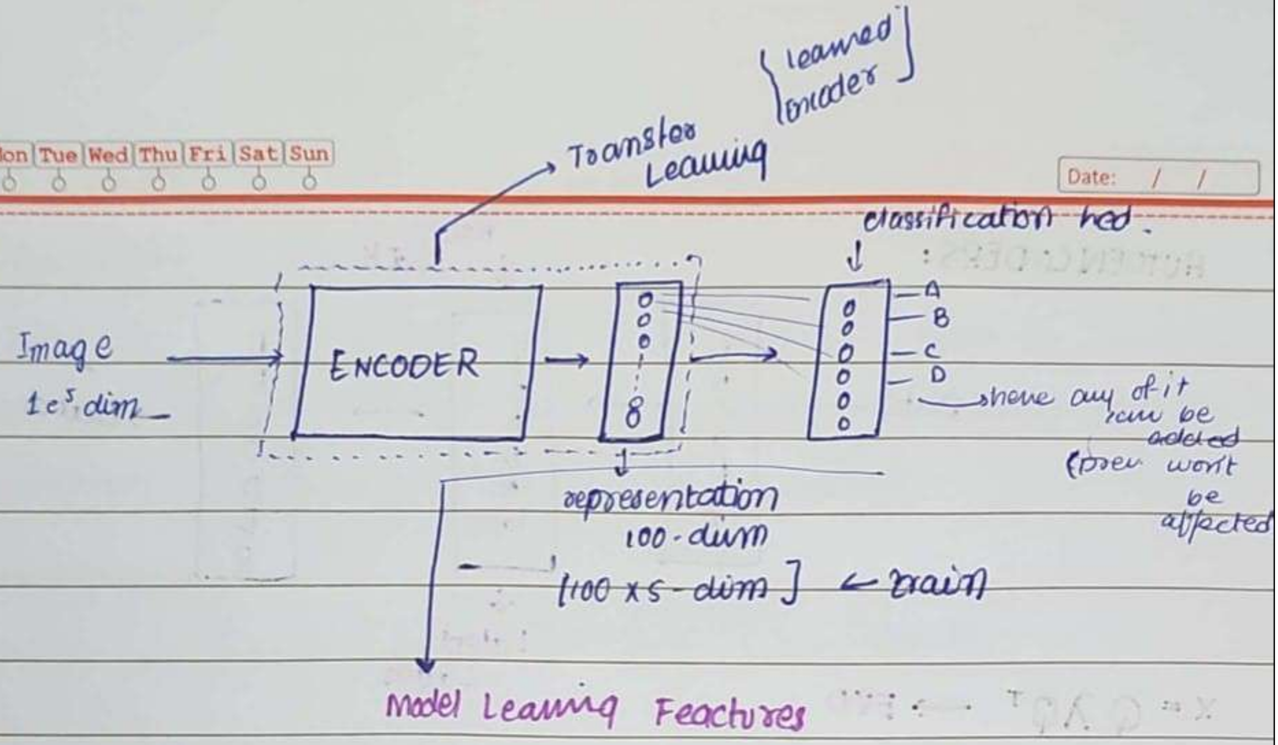
- $V : \mathbb{R}^P \longrightarrow \mathbb{R}^P$
  $\mathbb{R}^2$

SVD:

$\mathbb{R}^P \longrightarrow \mathbb{R}^K$ (measure in $\mathbb{R}^K$)
$(K \leq P)$

eg: $P = 4$, $K = 2$

$\begin{bmatrix} 5 & 2 & 0.1 & 0 \end{bmatrix}$




- Loss function promises Reconstruction, not generation.

## VARIATIONAL AUTOENCODERS:

- Image → Encoder → Latent Layer → Decoder → Probability Distribution P(x) → Images

  Images → Encoder → parameters of P(x).

- parameters → Decoder → Images

- N ~ (0,0) → Standard Gaussian Distribution $\mu = 0,\ \sigma^2 = 1$

Reconstruction Loss $= \|x - x^2_2\|^2$
measures distance between two vectors
✓ Entropy

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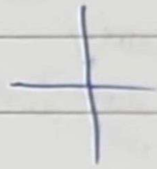
## GENERATIVE MODELS ⟷ Variational Autoencoders

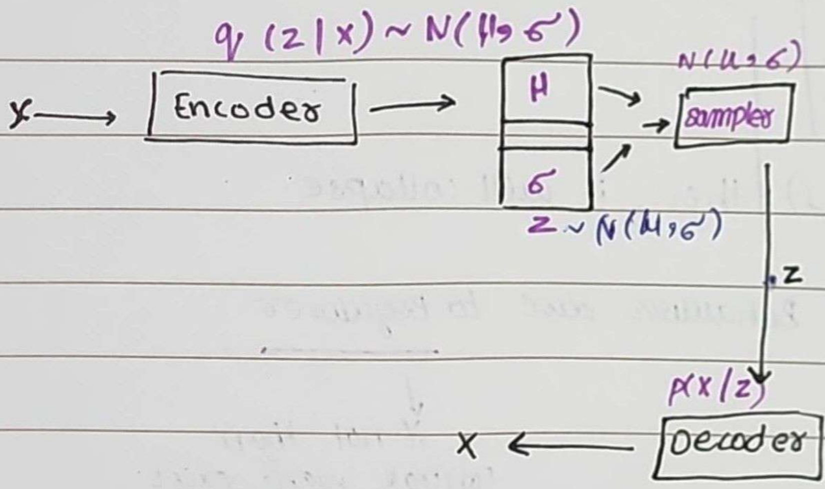
Recap: linear Algebra, Dimentionality Reduction (PCA, SVD, EVD), Autoencoders <Deterministic>

$\boxed{p(x)} \sim N(0,1)$ ← standered Guassian

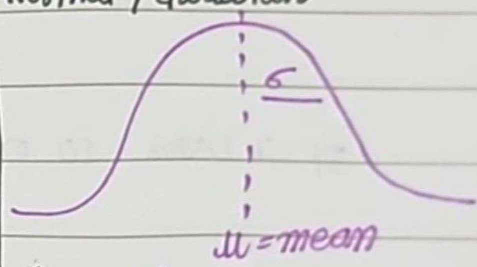
$f(x): \longrightarrow \mathbb{R}$



$q(z|x) \sim N(\mu, \sigma)$




Normal / Guassian




$N(\mu, \sigma^2)$

$$\frac{1}{2\pi\sigma^2} \cdot e^{\frac{(-x-\mu)^2}{2\sigma^2}}$$

$N(0,1) \rightarrow 1\, e^{-x^2/2}$

### LOSS - VARIATIONAL AUTOENCODERS

→ Divergence → measuring distance b/w two prability distributions.

$$\underset{(VAE)}{\text{LOSS}} = \underset{\text{loss}}{\text{Reconstruction}} + \underbrace{D_{KL}(q(z|x) \| p(x))}_{\text{REGULARIZER} \rightarrow p(x)}$$

$$E_{q(z|x)}[p(x|z)]$$

$$= \int q(z|x) \cdot p(x|z)\, dx$$

why 'distance'

$$\int q(z|x) \log \frac{q(z|x)}{p(x)}\, dx$$

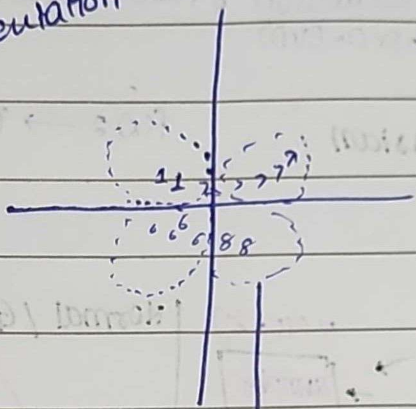
⇓

- distance of defined p(x)
- knowledge of another distribution in terms of know distribution.

Image → Representation: (Eg)

2D Representation

actually done it 1000-dims.




(Enforcing a behaviour making it learn ~~the~~ or recognize the patter)

If it learns (0 or 1) then, it will collapse.

Behaviour due to Regulizer

if not then controll won't exist.

mathematical terms used (Regularizer)

- Completeness
- Continuety